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Unit IV: Plant Anatomy (Structural organisation)

Chapter 9

Tissue and Tissue System

Learning Objectives

The learner will be able to,

- *Study major types of plant cells and their function.*
- *Differentiate the various types of cells.*
- *Study the relationship between the distribution of tissues in the various parts of plants.*
- *Describes the ground tissue system [cortex and pith] and vascular systems*
- *Interpret cross sections and longitudinal sections of dicot and monocot root, stem and leaf.*
- *Compare the internal organization of dicot root and monocot root.*

Nehemiah Grew
Father of Plant Anatomy



1641–1712

Katherine Esau (1898–1997)

A legendary Role model for women in science. She was a scintillating Botany teacher and pioneering researcher for six decades. Her classic book **Anatomy of Seed Plants** is the best literature in Plant Anatomy. In recognition of her distinguished service to science, she was awarded **National Medal of Science (1989) by USA.**



Chapter Outline

9.1 Meristematic tissue
9.2 Permanent tissues
9.3 The tissue system
9.4 Epidermal tissue system
9.5 Fundamental tissue system
9.6 Vascular tissue system
9.7 Comparision of primary structure

This chapter introduces the internal structure of higher Plants. The study of internal structure and organisation of plant is called plant Anatomy (*Gk: Ana = as under; temnein = to cut*). Plants have cells as the basic unit. The cells are organised into tissues. The tissues in turn are organised into organs. The different organs in a plant have different internal structures. It is studied by means of dissection and microscopic examination.